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8 JUNE 1959
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MILITARY SPECIFICATION

TIEDOWN, CARGO, AIRCRAFT, 5,000-LB CAPACITY,
TYPE MC-1

This specification has been approved by the Department of the Air Force and by the Navy Bureau of Aeronautics.

1. SCOPE

1.1 This specification covers one type of 5,000-pound capacity aircraft cargo tiedown, designated Type MC-1.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

Federal

QQ-P-416	Plating, Cadmium (Electrodeposited)
PPP-B-566	Boxes, Folding, Paperboard
PPP-B-585	Boxes; Wood, Wirebound
PPP-B-591	Boxes, Fiberboard, Wood-Cleated
PPP-B-601	Boxes, Wood, Cleated-Plywood
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner
PPP-B-636	Boxes, Fiber
PPP-B-676	Boxes, Set-Up, Paperboard
PPP-T-60	Tape; Pressure Sensitive Adhesive, Waterproof - for Packaging and Sealing

Military

JAN-P-100	Packaging and Packing for Overseas Shipment General Specification
MIL-P-116	Preservation, Methods of
MIL-B-138	Boxes, Wood, Fiberboard-Lined for Overseas Shipment (for Weight of Contents Not Exceeding 500 Pounds)
MIL-W-4088	Webbing, Textile, Woven, Nylon
MIL-B-4229	Boxes; Paperboard, Metal-Stayed
MIL-E-5272	Environmental Testing, Aeronautical and Associated Equipment, General Specification for

FSC 1670

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MIL-T-7807	Thread, Nylon
MIL-A-8625	Anodic Coatings, for Aluminum and Aluminum Alloys
MIL-B-10377	Box, Wood, Cleated, Veneer, Paper Overlaid
MIL-L-10547	Liners, Case, Waterproof
MIL-D-70327	Drawings, Engineering, and Associated Lists

STANDARDS

Federal

FED-STD-595	Colors

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U. S. Military Property

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Qualification.- The tiedown furnished under this specification shall be a product which has been tested and has passed the qualification tests specified herein.

3.2 Components.- The tiedown major components shall consist of nylon webbing and attachment-hook assemblies.

3.3 Selection of specifications and standards.- Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with Specification MIL-D-70327 except as provided in 3.3.1 and 3.3.2.

3.3.1 Commercial parts.- Commercial parts having suitable properties may be used where, on the date of invitation for bids, there are no suitable standard parts. In any case, commercial utility parts, such as snaps, rings, hooks, et cetera, having suitable properties may be used provided:

(a) They can be replaced by the standard parts (MS and AN) without alteration.

(b) The corresponding standard part numbers are referenced in the parts list and, if practicable, on the contractor's drawings.

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3.3.2 Standard parts.- With the exception in 3.3.1, MS and AN standard parts shall be used where they suit the purpose. They shall be identified on the drawings by their part numbers.

3.4 Design and construction.- The tiedown assembly shall be designed and constructed as an integral unit and shall not require attachment or assembly of separate parts during usage. The tiedown assembly shall be constructed of a single length of nylon webbing and shall include two attachment hooks, one of which shall incorporate a device for manually adjusting, tensioning, and releasing the webbing. The tiedown shall be so constructed that no parts will work loose in service. Dimensions shall be in accordance with figure 1.

3.4.1 The tiedown shall be so designed and constructed that it can be easily operated in confined areas and under extreme-temperature conditions by personnel wearing heavy gloves. Webbing shall not interfere with the operation and locking of the tensioning lever.

3.4.2 Webbing.- Webbing shall conform to Specification MIL-W-4088, type X, condition U, except that the webbing shall be latex-treated, and shall maintain 90 percent of its original strength after subjection to the abrasion test in accordance with Specification MIL-W-4088. All ends of the webbing shall be cut with a hot knife or seared to prevent fraying. The webbing shall be green, color 34087, in accordance with Standard FED-STD-595.

3.4.3 Sewing.- A stitching design is shown on figure 2; however, an alternate design may be selected by the manufacturer.

3.4.3.1 Cord.- Cord used in sewing the webbing shall conform to Specification MIL-T-7807, type I or II, class 1, minimum size No. 3.

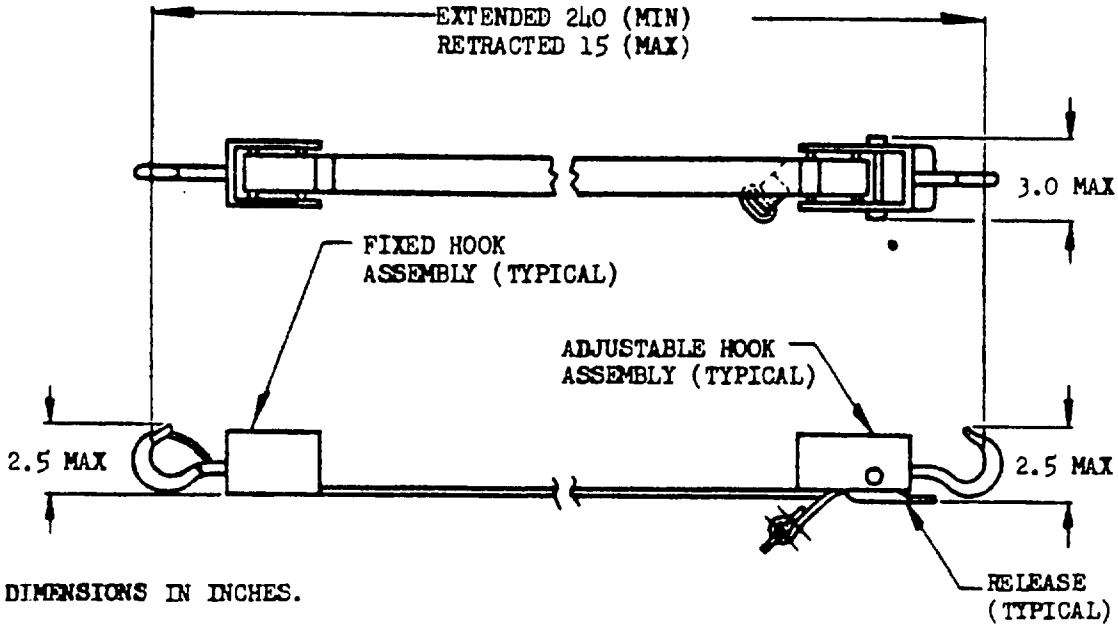
3.4.4 Attachment-hook assemblies.- A fixed-hook assembly and an adjustable-hook assembly shall be provided and designed for rapid attachment without special manual manipulation or use of tools. The hooks shall permit securing of the tiedown to a ring having a minimum clear opening of 3/4-inch diameter and a maximum cross-section diameter of 7/8 inch.

3.4.4.1 Fixed-hook assembly.- The fixed-hook assembly shall have a nonswivel hook with keeper. The envelope dimensions of the assembly shall not exceed those shown on figure 1.

3.4.4.2 Adjustable-hook assembly.- The adjustable-hook assembly shall have either a swivel or nonswivel hook, at the manufacturer's option, without keeper. The assembly shall incorporate an adjusting mechanism, a tensioning mechanism, and a release mechanism. Any mechanism of such design as to be hazardous to operating personnel will not be acceptable. The envelope dimensions of the assembly shall not exceed those shown on figures 1 and 2.

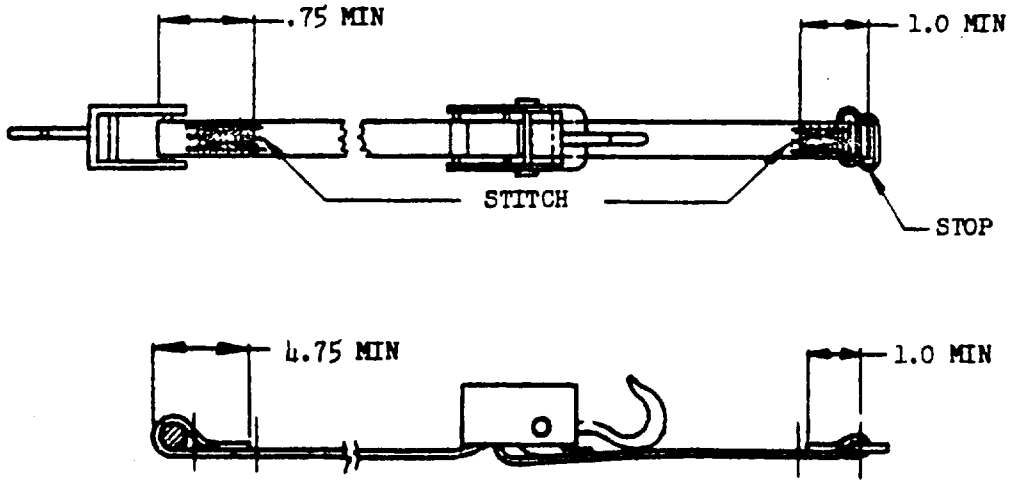
3.4.4.2.1 Adjusting mechanism.- The manually operated adjusting mechanism shall be capable of adjusting the tiedown to any lengths within the limits shown on figure 1. Adjustments in fixed increments of length shall not be acceptable. It shall be possible to accomplish preliminary adjustment of the webbing to the approximate length required in a maximum of 15 seconds.

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DIMENSIONS IN INCHES.

FIGURE 1. Envelope dimensions



DIMENSIONS IN INCHES.

FIGURE 2. Sewing

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3.4.4.2.2 Tensioning mechanism.- After preliminary adjustment of the webbing to remove excess slack, the tensioning mechanism shall be capable of applying, by manual operation, a tension to the webbing of not less than 150 pounds with an applied force of not more than 50 pounds. The tensioning mechanism shall be of either a single-action or a progressive-takeup ratchet-type tensioner. After tensioning, the mechanism lever shall not protrude thus causing a hazard to personnel moving about in the aircraft. If a ratchet-type mechanism is utilized, not more than six progressive takeups of the mechanism shall be required to obtain 150 pounds of tension and the mechanical advantage of the mechanism shall not exceed 3.5. The ratchet-type mechanism lever shall be of such design that it will not be possible to use a pipe or bar to increase the effective length of the lever and enhance the mechanical advantage of the mechanism. Surfaces which come in contact with the webbing shall not be knurled. A positive-type lock shall be incorporated in the tensioning mechanism and shall automatically engage upon completion of the tensioning operation. The force required to disengage the lock shall not exceed 5 pounds. An additional operation will be permitted to allow slack in the webbing (see 3.4.4.2.3).

3.4.4.2.3 Release mechanism.- Release of the tensioning mechanism shall introduce sufficient slack in the webbing to permit instant removal of the attachment hooks. The releasing force required to manually release a tension load of 225 pounds shall not exceed 75 pounds.

3.5 Performance.- The tiedown shall be capable of satisfactory operation under the following conditions:

(a) Exposure to salt sea atmosphere.

(b) Sand and dust particles as encountered in desert areas.

(c) Vibration incident to service usage.

(d) A temperature of -65° F.

3.5.1 Proof load.- The tiedown shall be capable of withstanding a proof load of 4,000 pounds for 30 seconds without evidence of failure or damage to webbing or sewing or permanent deformation of metal parts.

3.5.2 Ultimate load.- The tiedown shall be capable of withstanding an ultimate load of 5,500 pounds for 30 seconds without failure or slippage of the webbing or failure of sewing or metal parts.

3.6 Interchangeability.- All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The drawing number requirements of Specification MIL-D-70327 shall govern changes in the manufacturer's part numbers.

3.7 Weight.- The weight of the tiedown assembly shall not exceed 3.75 pounds.

3.8 Finishes and protective coatings.-

3.8.1 Steel.- Steel used in the tiedown shall be cadmium plated in accordance with Specification QQ-P-416.

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3.8.2 Aluminum alloy.- Aluminum-alloy parts used in the tiedown shall be anodized in accordance with Specification MIL-A-8625.

3.9 Identification of product.- Equipment, assemblies, and parts shall be marked for identification in accordance with Standard MIL-STD-130.

3.9.1 Use of AN or MIL designation.- Except for qualification test samples, AN or MIL designations shall not be applied to a product nor referred to in correspondence until notice of approval has been received from the activity responsible for qualification.

3.10 Workmanship.- Workmanship shall be of the highest quality to insure proper functioning under conditions to which the tiedown may be subjected. Unsatisfactory workmanship, such as loose, cocked or inadequately headed rivets, distorted or loose bushings and pins, loose, broken, or incomplete stitching, and rough, malformed, misalined, or improperly fabricated fittings shall be considered as defects. Corners shall be rounded, and sharp edges, burrs, or other protrusions that may damage the webbing shall not be permitted.

4. QUALITY ASSURANCE PROVISIONS

4.1 Unless otherwise specified herein, the supplier is responsible for the performance of all inspection requirements prior to submission for Government inspection and acceptance. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order.

4.2 Classification of tests.- The inspection and testing of tiedowns shall be classified as follows:

(a) Qualification tests (See 4.3.)

(b) Acceptance tests (See 4.4.)

4.3 Qualification tests.-

4.3.1 Qualification test samples.- The qualification test samples shall consist of two complete tiedowns. Samples shall be identified with the manufacturer's own part number and any additional information required by the letter of authorization.

4.3.2 Qualification required.- Prior to actual procurement, the tiedown which this specification covers shall pass the qualification tests specified herein. If the tiedown is modified in any way, the modified form shall be subjected to and shall pass the same qualification tests.

4.3.3 Tests.- Qualification tests shall consist of all the tests described under 4.6 and shall be conducted in the order listed. Both qualification test samples shall be subjected to the tests specified in 4.6.5 and 4.6.6.

4.4 Acceptance tests.- The acceptance tests shall consist of individual test and sampling plans A and B tests.

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4.4.1 Individual test.- Each tiedown shall be subjected to examination of product (4.6.1), as described under 4.6.

4.4.2 Sampling plans and tests.-

4.4.2.1 Lot.- The lot definition, formation, and size shall be in accordance with Standard MIL-STD-105.

4.4.2.1.1 Standard MIL-STD-105.- When Standard MIL-STD-105 specifies an action by the Government, it shall, at the option of the Government, be performed either by the Government or by the contractor under the supervision of the Government inspector.

4.4.2.2 Sampling plan A tests.- A sample of tiedowns shall be selected at random from each lot in accordance with Standard MIL-STD-105, inspection level I, and subjected to the following tests as described under 4.6. Acceptance and rejection shall be based on an AQL of 1.0.

(a) Tensioning test (4.6.2)

(b) Proof load test (4.6.5)

4.4.2.3 Sampling plan B tests.- One tiedown shall be selected at random from every one thousand or fraction thereof produced and subjected to the ultimate load test as described under 4.6. Sample tiedowns subjected to this test shall be in addition to the quantity specified in the contract and shall be furnished by the manufacturer without additional cost to the Government.

4.5 Test condition.-

4.5.1 Tiedown adjustment.- Unless otherwise specified, tests shall be made with the working length of the tiedown adjusted to 4 feet ±3 inches.

4.6 Test methods.-

4.6.1 Examination of product.- The tiedown shall be inspected to determine compliance with the requirements specified herein with respect to materials, workmanship, and marking.

4.6.2 Tensioning test.- The tensioning mechanism shall be operated until a tension of at least 150 pounds is applied to the webbing utilizing a maximum force of 50 pounds. The tiedown shall be tested and checked to determine compliance with all requirements of 3.4.4.2.2.

4.6.3 Release mechanism.- The tiedown shall be assembled in a testing machine and the release mechanism locked. A load shall be applied at a rate of separation of the heads of 1 inch per minute (under no load) until the testing machine indicates a load of 225 pounds. The release mechanism shall operate by the application of not more than 75 pounds force while the tiedown is sustaining the applied load. Upon release, the hooks shall be readily removable.

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4.6.4 Environmental tests.- The tiedown shall be subjected to the following tests in accordance with the applicable procedures of Specification MIL-E-5272.

4.6.4.1 Salt spray.- One tiedown shall be subjected to a salt-spray test in accordance with Procedure I. At the conclusion of this test, the tiedown shall be subjected to the tests specified in 4.6.2 and 4.6.3.

4.6.4.2 Sand and dust.- The tiedown specified in 4.6.4.1 shall be subjected to a sand and dust test in accordance with Procedure I. At the conclusion of this test, the tiedown shall be checked for compliance with 3.4.4.2.1 and subjected to the tests specified in 4.6.2 and 4.6.3.

4.6.4.3 Vibration.- The second tiedown shall be subjected to a vibration test in accordance with Procedure XII. Preceding this test, the tiedown shall be kept under a tension of 200 pounds for a 24-hour period. With the tiedown still under the 200-pound tension, the vibration test shall be conducted. No appreciable loss of tension shall result due to slipping of the tensioning mechanism.

4.6.4.4 Low temperature.- The tiedown specified in 4.6.4.3 shall be subjected to a low-temperature test in accordance with Procedure II. While at the temperature specified, the adjusting tensioning, and release mechanisms shall operate as specified in section 3.

4.6.5 Proof-load test.- The tiedowns shall be subjected to a proof load of 4,000 pounds for 30 seconds to determine compliance with the requirements of 3.5.1. During this test, there shall be no slippage of the webbing through the adjustment device.

4.6.6 Ultimate-load test.- The tiedowns shall be subjected to an ultimate load of 5,000 pounds for 30 seconds to determine compliance with the requirements of 3.5.2. During this test, damages such as a broken sewing thread without complete separation, broken or frayed webbing without complete separation of the webbing, and deformation but not rupture of hardware shall be permitted. The release mechanism shall be operable after being subjected to this test.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.-

5.1.1 Level A.- Tiedowns shall be preserved in accordance with Specification MIL-P-116, Method III, and packaged in containers conforming to Specifications PPP-B-566, PPP-B-636, PPP-B-676, or MIL-B-4229. Containers conforming to Specification PPP-B-636 may be used for domestic shipments. Unit quantities shall be as specified by the procuring activity.

5.1.2 Level C.- Tiedowns shall be preserved and packaged in accordance with the manufacturer's commercial practice.

5.2 Packing.-

5.2.1 Level A.- Tiedowns, preserved and packaged as specified in 5.1.1, shall be packed in exterior-type shipping containers conforming to Specifications

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PPP-B-585, PPP-B-591, PPP-B-601, PPP-B-621, PPP-B-636, MIL-B-138, or MIL-B-10377. Insofar as practicable, exterior containers shall be of uniform shape and size, of minimum cube and tare consistent with the protection required, and shall contain identical quantities. The gross weight of each pack shall be limited to approximately 200 pounds. Containers shall be closed and strapped in accordance with the applicable container specification or appendix thereto. Containers shall be provided with a case liner conforming to Specification MIL-L-10547 and shall be sealed in accordance with the appendix thereto. The case liner shall not be required when the unit, intermediate, or exterior container conforms to Specification PPP-B-636, and is sealed at all joints and seams, including manufacturer's joint, with tape conforming to Specification PPP-T-60.

5.2.2 Level B.- Tiedowns, preserved and packaged as specified in 5.1.1, shall be packed in domestic-type exterior containers conforming to Specifications PPP-B-585, PPP-B-591, PPP-B-601, PPP-B-621, PPP-B-636, or MIL-B-10377. Exterior containers shall be of minimum cube and tare consistent with the protection required. Insofar as practicable, exterior containers shall be of uniform shape and size and shall contain identical quantities. The gross weight of each pack shall be limited to approximately 500 pounds. Containers shall be closed and strapped in accordance with the applicable container specification or appendix thereto. When fiberboard containers are used, the fiberboard shall conform to Specification PPP-B-636.

5.2.3 Level C.- Packages which require overpacking for acceptance by the carrier shall be packed in exterior-type shipping containers in a manner that will insure safe transportation at the lowest rate to the point of delivery. Containers shall meet Consolidated Freight Classification Rules or regulations of other common carriers, as applicable to the mode of transportation.

5.2.4 Physical protection.- Cushioning, blocking, bracing, and bolting as required shall be in accordance with Specification JAN-P-100, except that waterproofing requirements for cushioning materials and containers shall be waived for domestic shipments. The drop test of Specification JAN-P-100 shall be waived when the item is preserved, packaged, and packed for immediate use or when the drop tests of Specification MIL-P-116 are applicable.

5.3 Marking of shipments.- Interior and exterior containers shall be marked in accordance with Standard MIL-STD-129. The shipment marking nomenclature shall be as follows:

TIEDOWN, CARGO, AIRCRAFT, 5,000-LB CAPACITY, TYPE MC-1

6. NOTES

6.1 Intended use.- The Type MC-1 tiedown covered by this specification is intended for use in lashing down cargo for transportation in military aircraft.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Selection of applicable levels of preservation, packaging, and packing.

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6.3 Qualification.- With respect to products requiring qualification, awards will be made only for such products as have, prior to the time set for opening of bids, been tested and approved for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government, tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Wright Air Development Center, Wright-Patterson Air Force Base, Ohio, and information pertaining to qualification of products may be obtained from that activity.

NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

Custodians:
Navy - Bureau of Aeronautics
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